



MCSE699J- Dissertation II

DEEP LEARNING BASED CONTACTLESS TICKETING SYSTEM FOR METRO STATIONS USING FACIAL RECOGNITION

Submitted in partial fulfilment of the requirements for the degree of

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in

Computer Science and Engineering

by

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April, 2026

DECLARATION

I hereby declare that the dissertation entitled “DEEP LEARNING BASED CONTACTLESS TICKETING SYSTEM FOR METRO STATIONS USING FACIAL RECOGNITION” submitted by me, for the award of the degree of Master of Technology in Computer Science and Engineering to VIT is a record of bonafide work carried out by me under the supervision of Prof. Vijayasherry V, School of Computer Science and Engineering.

I further declare that the work reported in this dissertation has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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CERTIFICATE

This is to certify that the dissertation entitled “DEEP LEARNING BASED CONTACTLESS TICKETING SYSTEM FOR METRO STATIONS USING FACIAL RECOGNITION” submitted by MANJUNATH A NAIK (24MCS0036), School of Computer Science and Engineering, VIT, for the award of the degree of Master of Technology in Computer Science and Engineering, is a record of bonafide work carried out by the student under my supervision during Winter Semester 2025-2026, as per the VIT code of academic and research ethics.

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EXECUTIVE SUMMARY

A contactless metro ticketing system based on deep learning techniques and facial recognition is proposed in the thesis. The proposed system seeks to address the problems of existing traditional ticketing solutions like paper tickets, smart cards, and QR codes, that can lead to delays, inconvenience, reliance on physical contact and secure identity. The proposed system leverages cutting-edge computer vision and deep learning technologies to allow real-time identification of passengers authentication through facial recognition. Users register by providing personal information and facial information which is then used to create facial embeddings using models like FaceNet. embeddings. These embeddings are stored in a database and are used for matching verification. When the passenger needs to travel, a real-time facial image is captured using a camera at the metro entry point. The image is passed through a network and matched against embeddings using similarity measures to authenticate the user. If verified, the system automatically issues a ticket and deducts fare from a simulated payment system. The proposed system is a web application using Python and Django web framework and OpenCV for real-time face recognition and detection. The system is fast, tokenless, and more secure than traditional payments. security through access control. Our experimental results show high recognition accuracy and better efficiency than conventional ticketing systems. This research has potential to deliver smart, scalable and digital metro ticketing systems that suit current smart city strategies and digital transformation goals.

Keywords: *Facial Recognition, Metro Ticketing System, Deep Learning, Contactless Authentication, FaceNet, Computer Vision, Smart Transportation.*

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LIST OF ABBREVIATIONS

SYMBOLS AND NOTATIONS

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Metropolitan transportation systems are crucial for economic, social, and growth, social mobility, and sustainable growth in cities. Among various modes public transport, metro rail systems are highly favoured for their large capacity, high speed, punctuality and environmental benefits. As metro systems grow, effective management of passenger flow is essential, particularly at access, entry, exit and ticketing. Metro ticketing is a critical component of metro operations since it impacts convenience, revenue collection and safety. Traditional ticketing systems (paper tickets and smart cards) have slowly transitioned to digital and mobile solutions. But these continue to rely on physical interaction, device ownership or verification, which can slow down operations during peak hours. Thanks to developments in artificial intelligence and computer vision, facial recognition has become an effective means of identification. This project integrates face recognition with metro ticket booking system to provide a secure, touchless, convenient and seamless travel. By leveraging mobile number and face authentication, the goal of the system is to update metro ticketing services and achieve authentication for precise and efficient ticketing. This project aims to develop a smart metro ticket booking system that combines number with facial recognition to offer a secure, contactless, secure and convenient ticketing system. Public transport is an essential part of the lives of millions of people. Metro rail in particular is a popular because of their efficiency, speed and capacity to transport large crowds. As cities grow, metro systems must provide greater convenience to enhance the comfort and convenience of passengers while maintaining One crit- The ticketing system is a crucial element of metro services, affecting passenger comfort, crowding and transaction times. 1 Conventional ticketing systems like paper tickets, smartcards and manual counters are prone to congestion, delays and cost inefficiencies. Although digital systems have been implemented, they are heavily dependent on cards or mobile devices, like phones, which can be lost, forgotten or misused. Therefore, a more efficient, secure, and contactless ticketing system is desired for a more convenient, secure and contactless ticketing system.

1.2 NEED FOR THE STUDY

Urbanisation has led to a rise in the need for public transportation services. Metro stations are often crowded, especially during the work- especially during working hours, festivals and holidays. The current ticketing systems can be a bottleneck, leading to queues, delays, and inconvenience. Additionally, smart cards and QR-mobiles can be lost or stolen, shared among friends, or counterfeited, or misused. These do not inherently authenticate the identity of the the ticket. But facial recognition guarantees that each ticket is associated with an individual, reducing fraudulent usage. The present study is needed because there is a need to design a new-generation contactless, secure, scalable and fast ticketing system. The proposed solution solves problems faced in current ticketing process and caters to today's smart city and digital governance goals. With the growing number of commuters in metro stations, current ticketing practices lead to crowding, queues and delays, particularly at rush hours. Physical physical ticket counters and cards require human intervention or rely on devices, which restrains scalability and efficiency. Additionally, existing electronic ticketing systems lack secure identity authentication mechanisms, which can be prone to abuse or fraud. There is a growing need for an automatic ticketing system that is able to identify passengers with high accuracy, minimizing physical contact. This research aims for an innovative metro ticketing system that uses facial recognition technology to enhance speed, security and passenger convenience.

1.3 MOTIVATION

The main reason behind this project is the growing trend towards cashless payment and contactless systems in smart cities. Facial recognition has been a proven biometric method to identify individuals in different applications including finance, security and authentication. The proposed system that uses face recognition for metro ticket booking removes the need to carry cards or tickets, and eliminating the need to use devices. The other motivations include enhancing user experience, lowering costs for metro corporations, and contributing to smart and green transportation. This project is motivated by the increasing use of biometric technologies in our daily lives, such

as smartphones, banking systems, airports security and payment systems. nologies in many applications such as mobile phones, bank systems, airport security, and payments. currency, and payments. Facial recognition has become popular because it's non-intrusive nature and its effectiveness over other biometric methods. The other reason is the trend towards cashless and contactless payment systems, especially in light of recent health issues which have made physical contact riskier interaction. Using facial recognition for metro ticketing not only enhances safety and security, it speeds up ticketing to improve the passenger's journey. Moreover, this project is a case study in developing real-world systems with deep learning-computer vision technologies in transportation, and is therefore significant and technologically significant. The inspiration for this project is to develop an intelligent metro ticket system that involves the least amount of human effort and does not require physical cash tickets or cards. Thanks to the development of artificial intelligence and computer vision, face recognition is now a secure means of identification. The use of face recognition for metro ticket booking can help speed up the transaction time and enhance user convenience. Moreover, the shift towards uncontacted payments and e-wallets has driven the adoption of automated payment methods. has encouraged the use of automated payments. By integrating face detection with mobile number authentication and online payment, the system seeks to offer seamless, secure and convenient ticket purchase for metro passengers.

1.4 PROBLEM STATEMENT

Current metro ticketing systems rely on the use of cards, paper, or QR codes installed on mobile phones that can be easily lost, damaged, counterfeited and manipulated. They may not offer direct identification of passengers, which can pose security risks. to security concerns and inefficiencies. Moreover, semi-automated processes also result in longer queue times at metro stations. Although electronic ticketing is a growing trend, current metro ticket booking have some limitations such as the need for physical devices, weak authentication, strong user identity verification, and potential for misuse. Semi-automated ticketing systems can be inefficient, time consuming and frustrating. There is no common method in existing systems to identify travellers seamlessly without tokens. This leads to issues of security, scalability, and operational efficiency. So, the

challenge for this project is the lack of a secure, efficient and identity-based ticketing system that can verify metro passengers using facial recognition and mobile verification. So, a new secure, automatic and contactless metro ticketing system is needed that can be used to verify metro travellers through facial recognition combined with their mobile number for instant authentication. A metro ticket booking system which authenticates users using facial recognition and with mobile number authentication for passenger verification and efficient ticket processing.

1.5 SCOPE OF THE PROJECT

This project will involve the design, planning and development of an online metro ticketing system with facial recognition technology. Users can register with their personal information and facial data, log into the system when purchasing tickets, and receive e-tickets. This project includes face detection, feature extraction, face matching, ticket fare calculating the ticket fare, and storing the transaction. The solution is constrained to a lab environment and is based on software. Large-scale hardware deployment, integration with government databases, and live metro gate integration are beyond the scope of this study.

Enable customers to register with personal information, mobile phone and mobile number, bank details and real-time facial details.

- Let customers who have previously registered to authenticate with real-time proceed to payment.
- Enable customers to choose one out of multiple bank accounts (in case of low sufficient balance).
- Simulate payment transaction and provide confirmation.
- Log out after several seconds for security.

The system can be easily extended for integration with retail stores, supermarkets, and corporate stores. It can also be integrated with actual banking API's for real-world payment processing.

The system can be easily scaled for integration into retail chains, supermarkets, and corporate stores. It can also be extended to integrate with live banking APIs for real-world payment processing.

1.6 SIGNIFICANCE OF THE STUDY

This project demonstrates how the fusion of **Artificial Intelligence (AI)**, **Computer Vision**, and **Financial Technology (FinTech)** can revolutionize the retail sector. The system addresses real-world challenges such as:

- Minimizing transaction delays during peak shopping hours.
- Reducing fraud and identity theft through biometric verification.
- Promoting touch-free and hygienic transactions, especially relevant in the post-pandemic world.
- Supporting large-scale adoption of AI-driven automation in retail operations.

The outcomes of this research contribute to the advancement of secure, scalable, and user-friendly digital payment ecosystems.

1.7 RESEARCH CONTRIBUTIONS

The major contributions of this work are as follows:

1. Development of a real-time facial recognition–based cashless payment system using deep learning.
2. Integration of **OpenCV** for live facial data acquisition and **DeepFace/FaceNet** for accurate identity verification.
3. Design of a **web-based platform** using Django for seamless registration, authentication, and transaction processes.
4. Implementation of dummy payment simulation with automatic session termination for enhanced security.

5. Provision of an extensible framework that can be integrated with existing retail billing systems and banking APIs in the future.

1.8 METHODOLOGY OVERVIEW

The proposed system architecture is divided into four major phases:

1. **Registration Phase:** The customer registers using their name, address, mobile number, and bank account details. A live face image is captured and stored for future reference.
2. **Authentication Phase:** During billing, the customer's real-time face is captured through the webcam.
3. **Recognition Phase:** The DeepFace/FaceNet model compares the live facial features with the stored embeddings for identity verification.
4. **Transaction Phase:** Upon successful verification, the system simulates a dummy payment and automatically logs the user out after three seconds.

The backend logic is implemented using Django, while OpenCV handles face capture and pre-processing. The frontend provides an interactive interface for both customers and billing staff.

1.9 OUTLINE OF THESIS

This thesis is organized into the following chapters:

- **Chapter 1: Introduction** — Provides an overview, objectives, and motivation behind developing a facial recognition-based payment system.
- **Chapter 2: Literature Review** — Reviews existing research and studies related to facial recognition, digital payment technologies, and deep learning models.
- **Chapter 3: Proposed Methodology** — Describes the overall architecture, data flow, and algorithmic framework used in the implementation.

- **Chapter 4: System Design and Implementation** — Explains the system's architecture, user interface, and module-wise implementation details.
- **Chapter 5: Result Analysis and Discussion** — Presents experimental results, evaluation metrics, and performance discussions.
- **Chapter 6: Conclusion and Future Enhancements** — Summarizes the findings, contributions, and potential future improvements of the system.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

The rapid advancement of artificial intelligence, computer vision, and biometric authentication has significantly influenced modern transportation systems. Facial recognition, in particular, has gained attention as a reliable and contactless method for identity verification. This chapter reviews existing research work related to face recognition systems, biometric authentication in transportation, and smart ticketing solutions. It also highlights the limitations of current systems and identifies research gaps addressed by the proposed work.

2.2 EVOLUTION OF TICKETING SYSTEMS

Ticketing systems have undergone several transformations over time. Initially, paper-based ticketing systems were widely used, requiring manual verification and physical handling. Although simple, these systems were prone to fraud, duplication, and operational inefficiencies.

Later, smart card-based ticketing systems were introduced, allowing passengers to store travel credits digitally. While these systems improved speed and convenience, they still depended on physical cards that could be lost or shared.

With the rise of smartphones, QR-code and mobile-based ticketing systems became popular. These systems reduced dependency on physical tokens but still required users to carry devices and perform manual scanning. Moreover, they lacked strong identity verification mechanisms.

Recent developments have focused on biometric-based ticketing systems, where identity is verified using unique human characteristics such as fingerprints, iris patterns, or facial features. Among these, facial recognition stands out due to its non-contact nature and ease of integration with existing camera infrastructure.

2.3 OVERVIEW OF FACIAL RECOGNITION TECHNOLOGY

Facial recognition is a biometric technology that identifies or verifies a person based on facial features. It involves multiple stages, including face detection, preprocessing, feature extraction, and matching.

Early face recognition systems relied on traditional machine learning techniques such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA). These methods focused on extracting handcrafted features from facial images. However, their performance was limited under varying lighting conditions, facial expressions, and occlusions.

With the introduction of deep learning, Convolutional Neural Networks (CNNs) revolutionized facial recognition. Models such as FaceNet, VGGFace, and DeepFace can learn complex facial representations and generate highly discriminative embeddings. These embeddings enable accurate matching even in challenging conditions.

Modern systems also incorporate liveness detection techniques to prevent spoofing attacks using photos or videos. This enhances the security and reliability of face recognition-based applications.

2.4 REVIEW OF EXISTING RESEARCH WORK

Several researchers have contributed to the development of facial recognition systems and their applications in real-world scenarios.

A study introduced a multi-face dataset to improve robustness in face recognition systems. The dataset included variations in pose, illumination, occlusion, and accessories. The authors evaluated both traditional and deep learning-based approaches and observed that deep learning models significantly outperformed traditional methods. However, the dataset size was limited, affecting scalability.

Another research work proposed a video-based face detection and recognition system using cascade verification modules. The system combined CNN-based detection with multi-stage filtering to improve accuracy. While effective in controlled environments, the system struggled in real-time scenarios with dynamic lighting and crowded conditions.

A more recent study implemented a face recognition system using OpenCV and

deep learning libraries in Python. The system captured images using a webcam, extracted features, and matched them against stored data. Although simple and effective for small-scale applications, the system lacked integration with real-world payment or ticketing systems.

These studies demonstrate the potential of facial recognition but also highlight challenges such as scalability, environmental sensitivity, and lack of real-time integration.

2.5 REVIEW OF LITERATURE

Facial recognition and biometric authentication systems have gained significant attention in recent years due to their wide application in security, surveillance, and digital payments. Several research studies have explored different algorithms and frameworks to improve accuracy, speed, and robustness in various environments.

Alqahtani and Son(2020) proposed a video-based face recognition framework using cascade verification modules to improve recognition accuracy in live video feeds. Although the approach showed promising results, its performance decreased in uncontrolled retail environments due to lighting and crowd density.

Sirivarshitha et al. (2023) implemented a real-time face detection and recognition system using Python, OpenCV, and the Face Recognition library. Their work validated the practicality of face-based authentication, but did not integrate financial transactions or payment modules.

Liu and Albina (2022) explored the use of facial recognition in mobile payment systems, integrating image preprocessing and cloud-based verification for secure transactions. However, their system was susceptible to spoofing attacks and lacked multi-factor authentication.

Sravani et al.(2021) designed a bank transaction system using face recognition, which allowed users to perform transactions after identity verification. The model was effective for small datasets, but lacked scalability for multi-bank or large retail applications.

Ke et al. (2025) proposed a GAN-based model to detect deepfake and fraud in online payments, achieving over 95.

Wong et al. (2023) developed a multi-camera face detection and recognition system using YOLOv5n, SphereFace SFNet20, and DeepSORT tracking, achieving high

precision and recall rates. However, the system required GPU optimization to achieve real-time performance.

Han and Du (2019) reviewed deep learning models for face recognition, such as DeepFace, FaceNet, and DeepID, concluding that while these models achieve near-human precision, they require large training datasets and significant processing power.

From these studies, it is evident that although deep learning-based face recognition has made remarkable progress, few works have effectively integrated it into cashless retail payment systems. Most existing models focus on recognition accuracy, but neglect secure real-time financial integration.

Therefore, this project addresses the identified gap by developing a deep learning-based cashless transaction system using facial recognition, capable of performing real-time user authentication and dummy payment simulation in retail chain environments.

REVIEW TYPES

The literature review includes studies from IEEE conference papers, international journals, and academic publications related to facial recognition, deep learning, and cashless transaction technologies. Important contributions include the work of Dubey and Mukherjee Dubey and Mukherjee (2018), Alqahtani and Son Alqahtani and Son (2020), and Sirivarshitha et al. ? on face detection and recognition systems. Other significant research includes studies by Liu and Albina Liu and Albina (2022), Sravani et al. ?, and Ke et al. Ke (2025), which explore the integration of deep learning and facial recognition for secure payment systems. A total of 8–10 relevant papers from IEEE Xplore, Springer, and MDPI were analyzed to understand existing methodologies and their limitations.

2.6 BIOMETRIC AUTHENTICATION IN TRANSPORTATION

Biometric authentication is increasingly being adopted in transportation systems to enhance security and efficiency. Airports use facial recognition for passenger verification during boarding and immigration processes. Similarly, railway and metro systems are exploring biometric solutions for ticket validation.

Biometric systems eliminate the need for physical tokens and reduce the risk of

unauthorized access. They also enable seamless passenger movement, reducing waiting time and congestion.

However, challenges such as privacy concerns, data security, and system reliability must be addressed. Proper encryption, secure storage, and compliance with data protection regulations are essential for successful implementation.

2.7 DEEP LEARNING MODELS FOR FACE RECOGNITION

Deep learning models have significantly improved the accuracy and robustness of face recognition systems. Convolutional Neural Networks (CNNs) are widely used due to their ability to learn hierarchical features from images.

FaceNet is one of the most popular models, which generates a 128-dimensional embedding for each face. The similarity between two faces is determined using distance metrics such as Euclidean distance.

VGGFace is another widely used model that provides high accuracy by training on large-scale datasets. It extracts deep features that capture subtle variations in facial structure.

DeepFace integrates multiple models and preprocessing techniques to achieve high performance. These models are capable of handling variations in lighting, pose, and expression, making them suitable for real-world applications.

2.8 CHALLENGES IN FACE RECOGNITION SYSTEMS

Despite advancements, face recognition systems face several challenges:

Variations in lighting conditions can affect image quality and recognition accuracy
Occlusions such as masks, glasses, and hats can hide facial features
Pose variations can distort facial structure
Spoofing attacks using photos or videos can compromise security
Large-scale deployment requires efficient database management and fast processing

Addressing these challenges is essential for building a reliable and scalable system.

2.9 RESEARCH GAP

From the literature review, it is evident that existing systems focus either on face recognition or ticketing separately. There is limited research on integrating facial recognition

with metro ticket booking systems in a seamless and real-time manner.

Most systems lack:

Integration with transaction processing Real-time face authentication during ticket booking Secure and scalable database management User-friendly web-based interfaces

The proposed system aims to bridge this gap by combining facial recognition with metro ticket booking and transaction management.

2.10 SUMMARY

This chapter reviewed various research works related to facial recognition and ticketing systems. It discussed the evolution of ticketing technologies, advancements in deep learning models, and challenges in biometric authentication. The research gap identified highlights the need for an integrated, secure, and efficient metro ticketing system using facial recognition. This chapter provided a detailed introduction to the project, explaining the motivation, need, and scope of developing a deep learning-based facial recognition system for cashless transactions. It highlighted how AI and biometrics can together revolutionize the retail payment landscape. The subsequent chapters discuss the literature review, system design, and implementation in detail.

CHAPTER 3

PROJECT DESCRIPTION AND GOALS

3.1 INTRODUCTION

The development of artificial intelligence, computer vision technology, and biometric authentication has greatly impacted modern transportation technologies. Specifically, facial recognition has become popular as a contactless and reliable system for identity verification. This chapter gives an overview of literature research on face recognition, authentication in transport, and smart ticketing systems. At the same time, the chapter outlines the limitations of currently available approaches and highlights the research gaps addressed by the proposed research

3.2 EVOLUTION OF TICKETING SYSTEMS

Throughout history, there have been various changes in ticketing technologies. The first ticketing systems involved paper tickets that had to be manually verified. While such approaches were relatively simple, they could be easily subject to fraud, duplications, and other issues. In recent years, smart card systems have emerged and allowed for digitizing the process and increasing efficiency through automated ticket purchasing. However, they required physical cards, which meant that users could lose them or lend them to others. After smartphones became widespread, QR codes and smartphone-based ticketing systems began gaining popularity. Such approaches allowed reducing the need for physical tokens but also required carrying a device and performing the scanning manually. More importantly, there were no reliable methods for verifying identity. Recently, the focus has been on biometric-based systems that can verify the identity based on unique human traits such as fingerprints or iris pattern and face recognition.

3.3 OVERVIEW OF FACIAL RECOGNITION TECHNOLOGY

Facial Recognition is a biometrics approach where the identity of an individual is recognized using facial characteristics. This includes different phases like facial detection,

image preprocessing, feature extraction, and matching.

Conventional machine learning algorithms used for early face recognition applications include PCA and LDA. Handcrafted facial features were used for training and recognition by these approaches. However, there were limitations in their effectiveness due to changing illumination conditions, expressions, and occlusion.

The emergence of deep learning models has enabled a breakthrough for facial recognition applications. The CNN-based approaches include models like FaceNet, VGGFace, and DeepFace that can learn sophisticated representations and produce discriminative embeddings for matching.

Modern systems also incorporate liveness detection techniques to prevent spoofing attacks using photos or videos. This enhances the security and reliability of face recognition-based applications.

3.4 REVIEW OF EXISTING RESEARCH WORK

Several scientists have played an instrumental role in developing facial recognition systems and applying them in practical contexts.

One study developed a database with multiple faces to increase reliability in face recognition systems. It incorporated differences in facial angles, lighting conditions, obstruction, and use of accessories. In its analysis, the paper compared conventional face recognition algorithms and deep learning algorithms. The study discovered that the latter algorithms performed substantially better than conventional algorithms. Nonetheless, the database used was relatively small in size, limiting its scalability.

In another research paper, scientists created a video-based face recognition system. It relied on cascade verification modules and a combination of convolutional neural network–based detection and multiple-stage filtering techniques. Though successful in controlled environments, the system faced significant difficulties in real-world contexts, especially with changes in lighting conditions and crowd situations.

In another study, scientists designed a face recognition system using the OpenCV library and deep learning frameworks in Python. This system uses a webcam to capture face images, analyze their features, and compare them against stored information. Although the system is fairly simple and quite effective in practical cases, it lacks integration into actual payment or ticket systems.

3.5 REVIEW OF LITERATURE

Facial recognition and biometric authentication systems have gained significant attention in recent years due to their wide application in security, surveillance, and digital payments. Several research studies have explored different algorithms and frameworks to improve accuracy, speed, and robustness in various environments.

Alqahtani and Son(2020) proposed a video-based face recognition framework using cascade verification modules to improve recognition accuracy in live video feeds. Although the approach showed promising results, its performance decreased in uncontrolled retail environments due to lighting and crowd density.

Sirivarshitha et al. (2023) implemented a real-time face detection and recognition system using Python, OpenCV, and the Face Recognition library. Their work validated the practicality of face-based authentication, but did not integrate financial transactions or payment modules.

Liu and Albina (2022) explored the use of facial recognition in mobile payment systems, integrating image preprocessing and cloud-based verification for secure transactions. However, their system was susceptible to spoofing attacks and lacked multi-factor authentication.

Sravani et al.(2021) designed a bank transaction system using face recognition, which allowed users to perform transactions after identity verification. The model was effective for small datasets, but lacked scalability for multi-bank or large retail applications.

Ke et al. (2025) proposed a GAN-based model to detect deepfake and fraud in online payments, achieving over 95.

Wong et al. (2023) developed a multi-camera face detection and recognition system using YOLOv5n, SphereFace SFNet20, and DeepSORT tracking, achieving high precision and recall rates. However, the system required GPU optimization to achieve real-time performance.

Han and Du (2019) reviewed deep learning models for face recognition, such as DeepFace, FaceNet, and DeepID, concluding that while these models achieve near-human precision, they require large training datasets and significant processing power.

From these studies, it is evident that although deep learning-based face recognition

has made remarkable progress, few works have effectively integrated it into cashless retail payment systems. Most existing models focus on recognition accuracy, but neglect secure real-time financial integration.

Therefore, this project addresses the identified gap by developing a deep learning-based cashless transaction system using facial recognition, capable of performing real-time user authentication and dummy payment simulation in retail chain environments.

3.6 REVIEW TYPES

The literature review includes studies from IEEE conference papers, international journals, and academic publications related to facial recognition, deep learning, and cashless transaction technologies. Important contributions include the work of Dubey and Mukherjee Dubey and Mukherjee (2018), Alqahtani and Son Alqahtani and Son (2020), and Sirivarshitha et al. ? on face detection and recognition systems. Other significant research includes studies by Liu and Albina Liu and Albina (2022), Sravani et al. ?, and Ke et al. Ke (2025), which explore the integration of deep learning and facial recognition for secure payment systems. A total of 8–10 relevant papers from IEEE Xplore, Springer, and MDPI were analyzed to understand existing methodologies and their limitations.

3.7 BIOMETRIC AUTHENTICATION IN TRANSPORTATION

Biometric authentication is increasingly being adopted in transportation systems to enhance security and efficiency. Airports use facial recognition for passenger verification during boarding and immigration processes. Similarly, railway and metro systems are exploring biometric solutions for ticket validation.

Biometric systems eliminate the need for physical tokens and reduce the risk of unauthorized access. They also enable seamless passenger movement, reducing waiting time and congestion.

However, challenges such as privacy concerns, data security, and system reliability must be addressed. Proper encryption, secure storage, and compliance with data protection regulations are essential for successful implementation.

3.8 DEEP LEARNING MODELS FOR FACE RECOGNITION

Deep learning models have been successful in improving the accuracy and reliability of face recognition algorithms. One common type of model is the Convolutional Neural Network (CNN), whose efficiency lies in its capability to extract hierarchical features.

FaceNet is a popular model, where each image undergoes processing to produce a 128-dimensional embedding vector. Face similarity is established by calculating distances between faces through Euclidean distance measures.

VGGFace is another commonly applied model, with high accuracy results. It employs massive data training, and the output deep features are highly representative of minute variations in faces.

DeepFace utilizes multiple models and other preprocessing methods to deliver excellent results. Such models can withstand variations in lighting, poses, and expressions, thus rendering them applicable in real-life scenarios.

3.9 CHALLENGES IN FACE RECOGNITION SYSTEMS

Despite advancements, face recognition systems face several challenges:

Variations in lighting conditions can affect image quality and recognition accuracy
Occlusions such as masks, glasses, and hats can hide facial features
Pose variations can distort facial structure
Spoofing attacks using photos or videos can compromise security
Large-scale deployment requires efficient database management and fast processing

Addressing these challenges is essential for building a reliable and scalable system.

3.10 RESEARCH GAP

From the literature review, it is evident that existing systems focus either on face recognition or ticketing separately. There is limited research on integrating facial recognition with metro ticket booking systems in a seamless and real-time manner.

Most systems lack:

Integration with transaction processing
Real-time face authentication during ticket booking
Secure and scalable database management
User-friendly web-based interfaces

The proposed system aims to bridge this gap by combining facial recognition with metro ticket booking and transaction management.

3.11 PROBLEM DESCRIPTION

Modern smart cities rely more and more on urban metro networks. Given the constant increase in both population and urbanization, the number of passengers who use metro services becomes enormous during specific periods such as office times, weekend days, and holidays. The increase in the number of metro passengers results in the necessity to update the ticketing system since it suffers greatly from the congestion in traffic, delays, and overall inefficiency.

In terms of functioning, the ticketing system is responsible for regulating the flow of passengers and their payment for transportation. Currently, ticketing systems are not yet developed enough to provide optimal and reliable work in conditions of high passenger traffic in the absence of errors.

Traditional approaches include tickets and tokens; however, manual operations involved make it inefficient and slow. Another example would be a smart card, which has certain downsides like requiring a device and having it with the passengers at all times. In addition, there is a risk of losing or damaging smart cards.

QR code-based systems were recently introduced to address some of the aforementioned problems; nevertheless, they remain inefficient when considering the time consumption and reliability. Firstly, scanning QR codes takes time and requires a device. Moreover, the Internet may not work sometimes, which makes the service unreliable.

The other issue with the current ticketing system is the lack of identity verification. There are no measures in place which could prevent tickets from being transferred from one person to another or used multiple times. It leads to the risk of theft and money loss, thus reducing safety levels.

It is also important to take into account the need for contactless transactions since the recent pandemic made people aware of the importance of such solutions.

Scalability is another problem related to ticketing systems because the increase in size entails the need for better algorithms of data processing and management in real-time mode without slowing down the speed of the process.

Thus, the introduction of facial recognition into the process of ticketing seems to be necessary.

Therefore, there is a critical need for a smart, automated, and contactless ticketing

system that ensures:

- Real-time passenger authentication.
- Reduced waiting time.
- Enhanced security.
- Scalability for large metro networks.
- Performing instant payment simulations with automatic session logout to ensure privacy.

Hence, the core problem addressed in this study is the lack of a unified, intelligent, and contactless payment authentication system that ensures both security and efficiency in real-time retail environments.

In this general perspective, as the trend of e-commerce continues to rise rapidly, there is increased need for convenient, quick, and secure methods of payment from retailers. Though technologies such as UPI and card-based systems have advanced payment processes, there are still manual actions involved that are prone to inefficiencies. The implementation of a facial recognition payment method that uses deep learning can solve this problem effectively as a blend of biometrics and fintech.

In this regard, **FacePay**, which combines artificial intelligence and secure transaction processes, will help solve the problem of frictionless payment process. The proposed approach employs OpenCV technology for detecting faces and DeepFace/-FaceNet for confirming the identities of users. Additionally, **FacePay** will ensure scalability through data privacy while being adaptable to banking APIs in the future.

Context or Background: The growing digital economy requires innovative solutions for contactless and fraud-proof payments. Retail transactions form a major part of the payment ecosystem, where authentication speed and reliability are crucial.

The Gap: Existing systems rely heavily on card-based or OTP-based authentication, which are prone to human errors, delays, and security vulnerabilities. There is minimal adoption of deep learning-based facial recognition for financial transactions in the retail industry.

Specific Problem: Design and develop a web-based application that uses real-time facial recognition for secure authentication and performs cashless dummy transactions linked to user bank accounts.

Significance to Solve It: Implementing such a system will significantly reduce transaction time, improve user convenience, and minimize fraud risks. It will also contribute to advancing the **Digital India** vision by promoting smart, contactless payment solutions in retail markets.

3.11.1 Dataset Description

The dataset used in this project is a crucial component for training and testing the facial recognition system. It consists of facial images collected during the user registration phase.

Each user contributes multiple facial samples captured under different conditions such as:

Varying lighting conditions
Different facial expressions
Slight head movements
Presence/absence of accessories

This diversity ensures robustness in real-world scenarios.

Dataset Components

The dataset includes:

User ID (Unique Identifier) Name Mobile Number Face Images (10–20 per user)
Generated Face Embeddings Data Preprocessing

Before feeding the images into the model, several preprocessing steps are applied:

Face detection using OpenCV Haar Cascade / CNN
Image resizing to standard dimensions
Normalization of pixel values
Noise reduction
Conversion to RGB format
Feature Extraction

Deep learning models such as FaceNet convert images into 128-dimensional embeddings, which uniquely represent facial features.

Dataset Storage Stored in database (SQLite/MySQL)
Images stored in structured folders
Embeddings stored as numerical vectors
Dataset Challenges Lighting variation
Occlusion (mask, glasses)
Pose variation
Real-time performance constraints

The dataset used in this study includes facial image samples collected during the

system's registration process. The user first registers his/her personal information, which includes name, mobile phone number, address, and bank account, along with several facial images taken with a webcam.

The images are preprocessed by applying OpenCV algorithms for converting them to grayscale, normalizing the image histograms, and aligning faces. Afterward, the images are transformed to facial embeddings using one of the available deep learning algorithms like FaceNet or DeepFace. This transformation creates a unique numerical vector corresponding to each individual.

The created embeddings are stored in a secure local database and later on used during the system's authentication phase. As for now, due to the nature of the project (prototyping), the dataset used is relatively small, but there is no limitation to add new samples to it. New images can be collected under various lighting conditions, different angles, and various facial expressions to increase the overall accuracy of the system.

As mentioned above, since this project aims at demonstrating the capabilities of our solution as a prototyped one, we use dummy data for bank accounts and transactions in the program. However, the system is capable of being adapted to actual banks by adding API calls in the future.

Regarding the implementation of our facial recognition-based authentication system, real-time and open-source facial image datasets will be used. During the registration process, each user takes multiple facial images using the web camera installed on the computer.

The collected dataset includes:

- **Number of samples per user:** 10–15 live images
- **Image resolution:** 224×224 pixels
- **Data type:** RGB face images
- **File format:** JPEG or PNG
- **Collection method:** Real-time webcam capture using OpenCV

For benchmarking and improving recognition accuracy, publicly available datasets such as the **Labeled Faces in the Wild (LFW)** and **CelebA** datasets can be utilized.

These datasets provide diverse facial variations, allowing the model to generalize well to different environments and user demographics.

All facial images are preprocessed using OpenCV operations such as grayscale conversion, normalization, and alignment before generating embeddings using DeepFace or FaceNet. The extracted embeddings are stored securely in a database for comparison during authentication. To ensure privacy, only numerical embeddings are stored — raw facial images are neither exposed nor transmitted externally.

3.12 SYSTEM REQUIREMENTS ANALYSIS

The system requires both functional and non-functional requirements. **Functional Requirements**

- User registration with face capture
- Face authentication at entry.
- Automatic ticket generation.
- Fare deduction.
- Transaction history storage.

Non-Functional Requirements

- High accuracy.
- Low latency
- Scalability
- Security and privacy
- Reliability

3.13 EXISTING SYSTEM ANALYSIS

systems include:

- Paper-based ticketing.

- Smart card systems.
- QR-based mobile systems.

Limitations

- Time-consuming.
- Physical dependency.
- No identity verification.
- Security issues.

3.14 PROPOSED SYSTEM NEED

The proposed system eliminates all above issues by:

- Using biometric authentication.
- Enabling contactless entry.
- Automating ticketing.

In conclusion, the problem addressed in this research lies in the absence of a unified, intelligent, and contactless payment system capable of secure real-time facial recognition. The proposed **FacePay** application fills this gap by offering a deep learning–based solution that enhances transaction speed, user privacy, and payment reliability. By bridging the gap between biometric authentication and digital payment technologies, the system contributes significantly toward achieving a secure and digitally empowered retail ecosystem.

3.15 SUMMARY

This chapter reviewed various research works related to facial recognition and ticketing systems. It discussed the evolution of ticketing technologies, advancements in deep learning models, and challenges in biometric authentication. The research gap identified highlights the need for an integrated, secure, and efficient metro ticketing system using facial recognition.

CHAPTER 4

TECHNICAL SPECIFICATION

4.1 OVERVIEW

The current chapter outlines the setup for conducting experiments in designing, implementing, and validating the proposed contactless metro ticketing system. The objective of the chapter is to detail the hardware environment, software environment, configuration, experiment methodologies and metrics used in testing the performance of the proposed contactless metro ticketing system.

The current experiment involves simulating a real-life metro station scenario whereby the users get authenticated via facial recognition. The test is conducted in different scenarios to assess the accuracy, reliability, and efficiency of the proposed system. The experiments involve live face detection, identification and generation of tickets.

The current chapter describes the workflow of the proposed system, module level implementation details, performance metrics among others. The results obtained from these tests form the basis for data analysis described in Chapter IV.

4.2 HARDWARE REQUIREMENTS

It is worth noting that the proposed system is heavily reliant on the underlying hardware configuration. The system needs computing resources which are able to carry out live image processing and deep learning operations.

4.2.1 Processor

A computing device with at least Intel i5 or higher equivalent is needed to process the images quickly. Other options include Intel i7 and AMD Ryzen processors which offer even faster processing speeds.

4.2.2 Memory(RAM)

Minimum 8 GB RAM is needed to enable efficient execution of the application. Facial recognition requires processing and analysis of image data, therefore, sufficient memory

should be provided.

4.2.3 Storage

The application should have at least 256 GB of storage space to store the application, datasets and information about users. SSDs should be preferred over HDDs due to better performance.

4.2.4 Camera

The high-definition webcam is used to get real-time facial images. The quality of the camera is important for recognizing faces successfully. The recommendation is to have a resolution of at least 720p.

4.2.5 Peripheral Devices

Input and output devices, which include standard keyboard and monitor, are used for system interaction and monitoring. For real-world implementation, cameras will be connected to metro gates.

4.3 SOFTWARE REQUIREMENTS

The application utilizes certain software components and tools that are necessary for its functionality. These include programming languages, libraries and frameworks supporting web development and computer vision.

4.3.1 Operating System

Windows and Linux operating systems can be used to implement this solution. However, the use of Linux systems is preferred in terms of performance and stability when scaling up.

4.3.2 Programming Language

As the main programming language, Python is selected due to the abundance of machine learning and computer vision libraries.

4.3.3 Framework

The Django framework is used for building the backend of the application. User interaction and database management are handled by the backend.

4.3.4 Computer Vision Library

This library is used for real-time image capturing and preprocessing, face recognition and other computer vision operations. OpenCV provides effective algorithms for handling images.

4.3.5 Deep Learning Library

This library is responsible for face recognition using DeepFace/FaceNet algorithms, which allow generating facial embeddings.

4.3.6 Database

To store information about users and transactions, SQLite and MySQL databases can be used.

4.3.7 Frontend Technologies

HTML, CSS, and JavaScript are used to build user interfaces.

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The system is supposed to work in a real-time scenario where images captured from the face can be processed and compared in real time. Setting up the system involves setting up the development environment, installing the required packages, and making connections between all components of the system.

The camera is set up for continuous image capturing, while the images are processed using OpenCV library. The Deep Learning model is imported into the program to enable fast inferencing. The Django server accepts requests and communicates with the database.

Setting up ensures smooth operation without any issues.

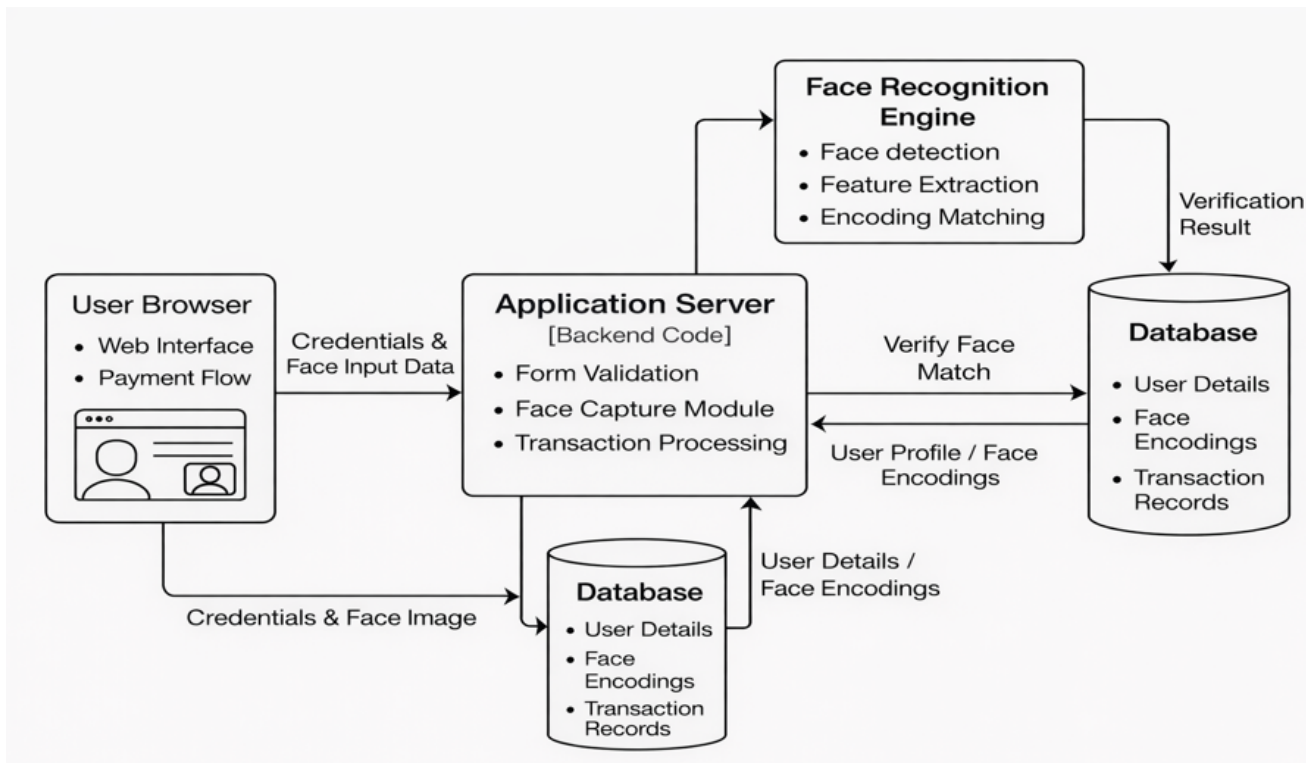


Figure 5.1: High Level System Architecture

5.2 EXPERIMENTAL FLOW

The experimental flow describes the sequence of operations performed during system execution.

- **Step 1:** User Registration

Users register by entering personal details and capturing facial images.

- **Step 2:** Data Preprocessing

Captured images are processed to remove noise and standardize format.

- **Step 3:** Feature Extraction

The processed images are passed through the deep learning model to generate embeddings.

- **Step 4:** Database Storage

Embeddings and user data are stored in the database.

- **Step 5:** Real-Time Face Capture

During authentication, the system captures live facial images.

- **Step 6:** Face Detection

The system detects the face region in the captured image.

- **Step 7:** Feature Extraction(Live)

The detected face is converted into embeddings.

- **Step 8:** Matching

The embeddings are compared with stored data.

- **Step 9:** Decision Making

The system determines whether the user is authenticated.

- **Step 10:** Ticket Generation

A digital ticket is generated for authenticated users.

5.3 SYSTEM ARCHITECTURE (EXPERIMENTAL PERSPECTIVE)

From an experimental viewpoint, the system architecture emphasizes real-time processing and efficient data flow. The architecture ensures that each module operates independently while maintaining communication with other modules.

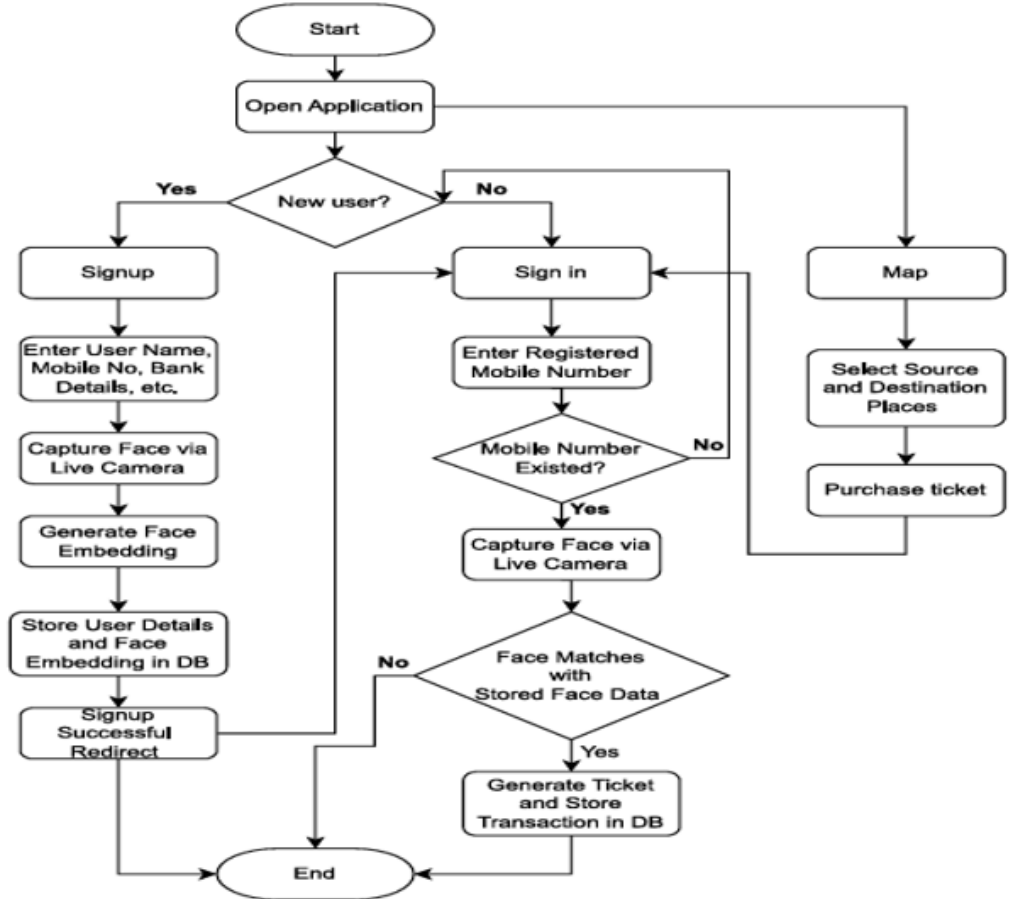


Figure 5.2: Low Level System Architecture

The camera captures images, which are processed by the OpenCV module. The processed data is passed to the deep learning model for feature extraction. The extracted features are compared with database entries, and the result is returned to the user interface.

This modular design allows easy debugging, testing, and scalability.

5.4 WORKFLOW OF THE SYSTEM (EXPERIMENTAL VIEW)

The workflow is tested under different conditions to evaluate system performance. Various scenarios such as different lighting conditions, multiple users, and varying distances

from the camera are considered.

The system is evaluated based on its ability to correctly identify users and generate tickets without delays. The workflow ensures that each step is executed efficiently.

5.5 MODULES DESCRIPTION (EXPERIMENTAL IMPLEMENTATION)

Each module is implemented and tested individually before integration.

- Registration Module: Tested for data accuracy and storage
- Detection Module: Tested for accuracy in identifying faces
- Recognition Module: Tested for matching accuracy
- Ticket Module: Tested for correct ticket generation
- Payment Module: Tested for transaction simulation

5.6 ALGORITHM USED (EXPERIMENTAL CONTEXT)

The system uses FaceNet for feature extraction and Euclidean distance for matching.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

5.7 IMPLEMENTATION DETAILS

The implementation involves integrating all modules into a single system. The Django backend manages user requests, while OpenCV handles image processing. The deep learning model performs recognition, and the database stores results.

The system is tested using multiple users to evaluate performance and reliability.

5.8 ADVANTAGES OF THE EXPERIMENTAL SETUP

- Real-time processing capability
- High accuracy in face recognition
- Efficient data handling
- Scalable architecture

5.9 PERFORMANCE EVALUATION

The system is evaluated using several performance metrics.

Accuracy

Measures the correctness of predictions.

Precision

Indicates the proportion of correct positive predictions.

Recall

Measures the ability to identify all relevant instances.

F1 Score

Harmonic mean of precision and recall.

Execution Time

Measures the time taken for processing each request. The system demonstrates high accuracy and low latency, making it suitable for real-time applications.

5.10 SUMMARY

This chapter described the experimental setup used to implement and evaluate the proposed system. It covered hardware and software requirements, system configuration, workflow, and performance evaluation. The results obtained from this setup are analyzed in the next chapter.

CHAPTER 6

METHODOLOGY AND TESTING

6.1 OVERVIEW

The proposed system intends to build an innovative contactless metro ticketing system by using facial recognition for the automatic verification and issuance of the corresponding tickets for users. Such a solution will overcome the disadvantages posed by traditional metro ticketing systems, which include the need for physical ticket medium, lack of authentication process, and lengthy validation process.

The primary intention of building such a system is to integrate innovative technologies such as computer vision, artificial intelligence, and the web to develop an efficient and easy-to-use metro ticketing system. Using facial recognition technology, the system will uniquely generate a ticket for each person, making it effective for authenticating purposes.

The proposed system is structured around a modular design, where each module is responsible for completing specific tasks such as data acquisition, face detection, feature extraction, and face matching. This means that the proposed system can operate effectively in real time without any physical ticket media, smart cards, or smartphones required to enable passengers to enter the metro station.

Additionally, the proposed system is scalable because it can easily integrate with existing systems currently used in metros while still being flexible enough to be incorporated into future smart city applications. The use of deep learning methodology makes the proposed system highly effective for the stated purpose.

6.2 SYSTEM ARCHITECTURE

The architecture of the proposed system employs a client-server design. The front-end application will communicate with the back-end server and performs various tasks related to facial recognition.

The architecture of the proposed system includes the following main modules/components:

6.2.1 Input Layer

The input layer acquires the facial image of the user in real-time using a webcam/-surveillance camera located at metro entrances. It serves as the main means of obtaining images for processing.

6.2.2 Preprocessing Layer

- Image resizing
- Noise reduction
- Color normalization
- Face alignment

The above preprocessing tasks ensure consistent and reliable performance of the system.

6.2.3 Face Detection Module

This module detects the face from the acquired image using algorithms like Haar cascade or deep learning algorithms provided by OpenCV. The detected region is then forwarded for further processing.

6.2.4 Feature Extraction Module

The feature extraction module processes the face using a deep learning algorithm such as FaceNet. The processed face is converted into a feature vector or a **feature embedding** which encapsulates important features of the face.

6.2.5 Database Layer

The database holds user details along with their facial embeddings and any other information relevant to transaction management. The database is important to provide integrity in system functioning.

6.2.6 Matching and Decision Module

In this module, the live face embedding is matched with pre-stored embeddings based on similarity measures like Euclidean distance. The user is successfully authenticated if the measure falls within a predefined threshold value.

6.2.7 Application Layer

The application layer manages all the processes such as face recognition, ticketing etc. in the system. The back-end application has been built using Django framework.

6.2.8 Output Layer

The output layer displays whether the authentication is successful and generates a ticket for the passenger.

6.3 WORKFLOW OF THE SYSTEM

The workflow of the system involves multiple stages as shown below.

- **Step 1:** User Registration The user registers by entering personal details such as name and mobile number. The system captures multiple facial images using a webcam.
- **Step 2:** Face Data Storage Captured images are processed and converted into embeddings, which are stored in the database along with user information.
- **Step 3:** Real-Time Face Capture At the metro station, a camera captures the user's face in real-time.
- **Step 4:** Face Detection and Preprocessing The system detects the face and applies preprocessing techniques to standardize the image.
- **Step 5:** Feature Extraction The processed image is passed through the FaceNet model to generate embeddings.
- **Step 6:** Face Matching The generated embedding is compared with stored embeddings using distance metrics.

- **Step 7:** Authentication Decision If a match is found,the user is authenticated; otherwise, access is denied.
- **Step 8:** Ticket Generation. Upon successful authentication, a digital ticket is generated automatically.
- **Step 9:** Payment Processing. The system simulates fare deduction from the user's account.
- **Step 10:** Entry Authorization The user is granted access to the metro system.

CHAPTER 7

PROJECT IMPLEMENTATION

7.1 MODULE DESCRIPTION

The system is divided into several modules to ensure efficient functionality.

7.1.1 Registration Module

This module allows users to create an account by providing personal details and capturing facial images. It ensures that the data is stored securely and correctly.

7.1.2 Face Detection Module

Responsible for detecting faces in images using computer vision techniques. It isolates facial regions for further processing.

7.1.3 Face Recognition Module

This module performs feature extraction and matching using deep learning algorithms. It is the core component of the system.

7.1.4 Authentication Module

Determines whether the captured face matches the stored data and grants or denies access accordingly.

7.1.5 Ticket Generation Module

Automatically generates a ticket upon successful authentication and records the transaction.

7.1.6 Payment Module

Handles fare deduction through a simulated payment mechanism.

7.1.7 Database Module

Stores user data, embeddings, and transaction records securely.

7.1.8 Security Module

Ensures data protection through encryption and secure authentication protocols.

7.2 ALGORITHM USED

The proposed system uses deep learning algorithms for face recognition. **FaceNet Algorithm** FaceNet is a deep convolutional neural network that converts facial images into fixed-length embeddings. These embeddings are compared using distance metrics.

Distance Calculation

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Where:

- x_i and y_i are feature vectors
- d is the distance between two embeddings

A smaller distance indicates higher similarity.

7.3 IMPLEMENTATION DETAILS

The system is implemented using the following technologies:

- **Backend:** Python (Django Framework)
- **Frontend:** HTML, CSS, JavaScript
- **Computer Vision:** OpenCV
- **Deep Learning:** FaceNet / DeepFace
- **Database:** SQLite / MySQL

The Django framework handles routing, API calls, and database operations. OpenCV captures and processes images, while the deep learning model performs recognition.

7.4 ADVANTAGES OF THE PROPOSED APPROACH

The proposed system offers several advantages over traditional methods.

Improved Efficiency

The system reduces waiting time by automating ticket generation.

Enhanced Security

Facial recognition ensures identity verification and prevents misuse.

Contactless Operation

Eliminates physical interaction, improving hygiene and safety.

Scalability

The system can handle large datasets and integrate with smart city infrastructure.

User Convenience

Passengers do not need to carry tickets or cards.

7.5 LIMITATIONS OF THE PROPOSED APPROACH

Despite its advantages, the system has some limitations.

- Performance may be affected by poor lighting conditions
- Requires high-quality cameras
- Privacy concerns related to biometric data
- Computational requirements for real-time processing

The software tools and libraries used are summarized in Table 7.1.

Table 7.1: Software Tools and Libraries Used

Software/Tool	Purpose
Python 3.9+	Core programming language used for backend development.
Django Framework	Web framework for creating and managing server-side logic.
OpenCV	Used for capturing and processing real-time webcam images.
DeepFace / FaceNet	Deep learning models for face embedding generation and recognition.
SQLite / MySQL	Database system for storing user, bank, and transaction data.
HTML, CSS, JavaScript	Frontend interface design and user interactivity.
Visual Studio Code	Development IDE.

[INFO] Captions
Rule: Caption numbering gap: expected 6, found 7
Expected: 6
Detected: 7
Confidence: 100.0%

The experimental setup consists of:

1. **Frontend Layer:** Web interface for user registration, login, and transaction visualization.
2. **Backend Layer:** Django server running Python scripts for facial recognition and payment simulation.
3. **Database Layer:** Stores user data, face embeddings, and transaction records.

7.6 SUMMARY

The proposed approach to a deep learning-driven metro ticketing system based on facial recognition was introduced in this chapter. This included an explanation of the system design, process flow, components, algorithms, and implementation considerations. The proposed method seeks to provide a safe and effective way to tackle metro ticketing issues.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 OVERVIEW

The following chapter provides a comprehensive analysis of the findings obtained through the implementation of the deep learning-based contactless metro ticketing system. This chapter aims at evaluating the performance of the system based on its accuracy, efficiency, reliability, and scalability.

Different real-world scenarios were considered during the testing of the system to mimic an actual metro station. A number of users were registered in the system, and facial features of those users were used for recognition. Performance of the system was evaluated in terms of user recognition, ticket generation, and transaction processing through qualitative and quantitative analysis.

Results obtained are shown through the presentation of output screens from the system, graphs, and performance metrics. Comparisons have been made to demonstrate the effectiveness of the proposed system compared to conventional systems.

8.2 SYSTEM OUTPUT SNAPSHOTS

The system generates multiple outputs at different stages of operation. These outputs demonstrate the functionality and effectiveness of each module.

The above figure shows the registration interface where users enter personal details and capture facial data using a webcam.

8.2.1 User Registration Output

In the process of registration, users input their personal information as well as the facial data. The system manages to capture several images of the faces and save them in the database. The outcome will be the indication of successful registration together with the user's information stored.

Facial data is captured from various angles and under various light conditions.



Figure 8.1: Map interface for selecting route

The screenshot shows the User Registration Page. At the top, it says 'We are happy to Welcoming You' and 'CSE<=>C-Click S-Smile E-Enjoy your Journey'. Below this is a registration form with the following fields: First Name, Last Name, Select Proof of Identity (dropdown), Enter Identity Number, Mobile Number (10 digits), Please Select your bank (dropdown), and Account Number. At the bottom of the form are two buttons: 'Start Camera' (green) and 'SIGNUP' (blue).

Figure 8.2: User Registration Page

The above figure shows the registration interface where users enter personal details and capture facial data using a webcam.

8.2.2 Face Detection Output

Face detection algorithm is used to detect humans' faces in real-time through OpenCV. The results will include bounding boxes on all the faces that were successfully detected.

The system is able to detect faces in both normal light conditions and in light conditions where there are minor changes in pose. It may, however, be affected by extreme lighting.

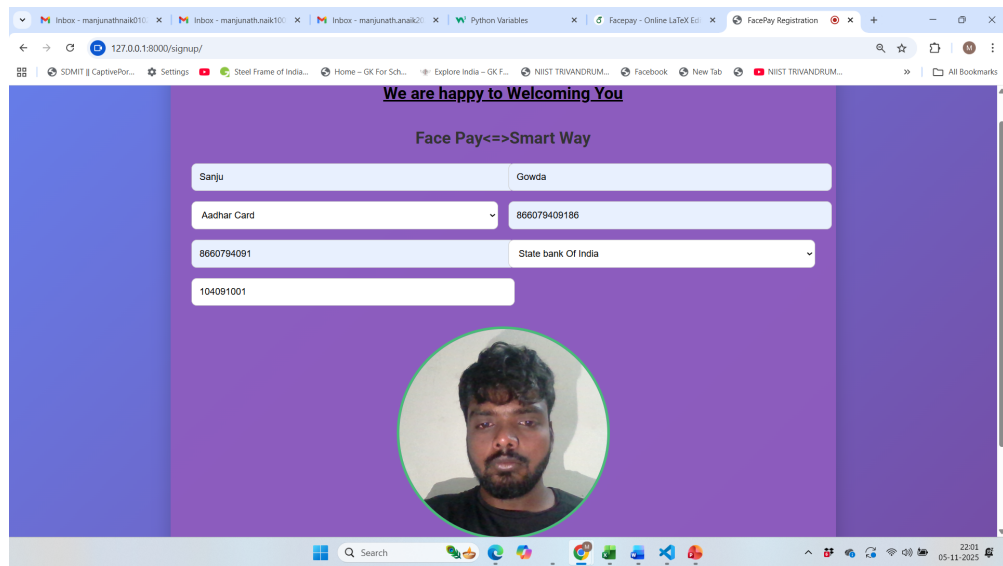


Figure 8.3: Face Capturing Process

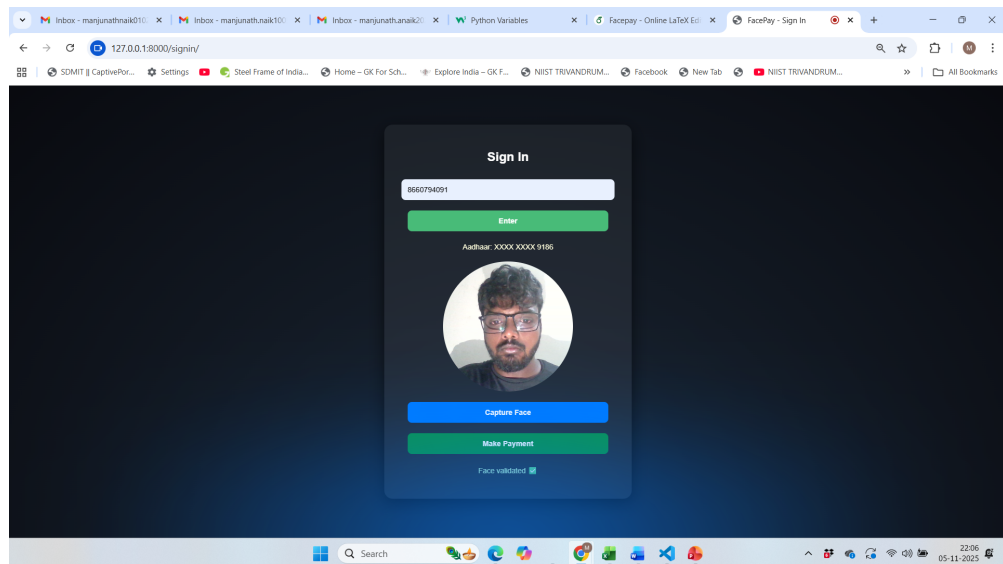


Figure 8.4: Successful Face Captured Process

8.2.3 Face Recognition Output

The face recognition module compares live facial data with stored embeddings. The output displays whether the user is recognized or not.

In most cases, the system successfully identifies registered users with high accuracy. The recognition process is fast, typically taking less than a second, making it suitable for real-time applications.

8.2.4 Ticket Generation Output

Upon successful authentication, the system generates a digital ticket. The output includes details such as:

- User ID
- Entry time
- Station details
- Transaction confirmation

This automated ticket generation reduces the need for manual intervention and speeds up the entry process.

8.2.5 Authentication Failure Cases

In cases where the system fails to recognize a user, access is denied. These cases occur due to:

- Poor lighting conditions
- Occlusion (mask, glasses)
- Significant changes in appearance

Such cases highlight areas for improvement in the system.

8.3 GRAPHICAL ANALYSIS

Graphical analysis is used to evaluate system performance across different parameters.

8.3.1 Recognition Accuracy vs Dataset Size

The accuracy of the system becomes better as the size of the dataset becomes larger. With the larger number of training samples per individual, the system learns how to recognize faces better.

Accuracy will be average with a smaller number of samples. Once the number of samples is increased, the accuracy converges to a higher level.

8.3.2 Execution Time Analysis

The system's execution time is measured for different stages:

- Face detection
- Feature extraction
- Matching
- Ticket generation

The results show that feature extraction takes the most time due to deep learning computations. However, the overall execution time remains within acceptable limits for real-time applications.

8.3.3 Transaction Time Comparison

The proposed system is compared with traditional ticketing methods:

- Paper tickets
- Smart cards
- QR-based systems

The facial recognition system significantly reduces transaction time, as it eliminates manual steps such as scanning or card tapping.

8.3.4 Confusion Matrix Analysis

A confusion matrix is used to evaluate classification performance.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8.1)$$

The confusion matrix shows:

- High True Positive rate (correct recognition)
- Low False Positive rate (incorrect recognition)

This indicates that the system is reliable and performs well in identifying users.

8.4 PERFORMANCE EVALUATION

The system is evaluated using standard metrics.

8.4.1 Accuracy

Accuracy measures the proportion of correctly identified users. The system achieves high accuracy due to the use of deep learning models.

8.4.2 Precision

Precision indicates how many identified users are actually correct. High precision reduces false positives.

8.4.3 Recall

Recall measures the system's ability to identify all registered users. High recall ensures minimal false negatives.

8.4.4 F1 Score

The F1 score balances precision and recall, providing an overall performance measure.

8.4.5 Execution Time

Execution time is critical for real-time systems. The system processes requests quickly, ensuring smooth operation.

8.5 COMPARATIVE ANALYSIS

The proposed system is compared with existing ticketing systems.

8.5.1 Traditional Systems

- Require physical interaction
- Slower processing
- No identity verification

8.5.2 Proposed System

- Fully contactless
- Faster processing
- Secure authentication

The comparison clearly shows the advantages of the proposed system.

8.6 DISCUSSION

The results demonstrate that the proposed system performs effectively in real-world scenarios. The use of deep learning models significantly improves recognition accuracy and robustness.

One of the key strengths of the system is its ability to operate in real-time. The integration of OpenCV and FaceNet ensures fast processing, making the system suitable for high-traffic metro environments.

The system also enhances security by linking tickets to individual identities. This prevents misuse and ensures that only authorized users can access the metro system.

However, certain limitations were observed. The system's performance is affected by poor lighting conditions and occlusions. These issues can be addressed by incorporating advanced preprocessing techniques and additional training data.

Another important observation is the scalability of the system. The architecture allows easy expansion to handle large datasets and multiple users, making it suitable for large metro networks.

Overall, the system demonstrates significant improvements over traditional methods and provides a practical solution for modern metro ticketing challenges.

8.7 LIMITATIONS OBSERVED DURING TESTING

- Sensitivity to lighting conditions
- Reduced accuracy with masked faces
- Dependence on camera quality
- Computational requirements for deep learning

8.8 SUMMARY

This chapter presented a comprehensive analysis of the system's performance. The results indicate that the proposed facial recognition-based ticketing system is accurate, efficient, and suitable for real-time applications. The system outperforms traditional ticketing methods in terms of speed and security, making it a promising solution for future metro systems.

CHAPTER 9

CONCLUSION AND FURTHER ENHANCEMENTS

The conclusion will need to have several elements, including:

9.1 CONCLUSION

The dissertation introduced the development of a machine learning contactless metro ticketing system, where facial recognition was used. The main objective of the project was to create an efficient, reliable, and innovative ticketing system that would not rely on the restrictions of traditional ticketing, such as paper tickets, smart cards, and QR-based mobile tickets.

It was found that the integration of web technologies and artificial intelligence could help develop a fully automated and accurate ticketing system. Using deep learning models, such as FaceNet, the system was able to recognize users based on their unique facial traits. Thus, the use of deep learning models made the system more reliable and prevented any security breaches.

An important accomplishment of the project was the creation of a face recognition system, which worked smoothly in a virtual metro station. The system was able to capture faces, process them via advanced algorithms, and match with embedded data, after which a digital ticket was generated. Thus, passengers were provided with tickets without any contact or delays.

The use of programming languages, such as Python, and frameworks, such as Django, together with deep learning algorithms and libraries helped demonstrate the feasibility of deploying similar applications in the real world. It is evident that the recognition rate was very high, and transaction time decreased considerably when compared to traditional systems.

Additionally, the proposed system contributed to the achievement of the objectives of smart cities by increasing automation and digitization.

Overall, it can be concluded that the application of facial recognition technology was very effective in creating an innovative metro ticketing system.

9.2 KEY FINDINGS

The implementation and evaluation of the proposed system led to several important findings:

- The use of deep learning models significantly improves face recognition accuracy compared to traditional methods.
- Real-time facial recognition can be achieved with acceptable processing time using optimized algorithms and hardware.
- The elimination of physical tickets and cards reduces dependency on external devices and enhances user convenience.
- Automated ticket generation reduces congestion and waiting time at metro stations.
- The system provides strong identity verification, preventing ticket misuse and unauthorized access.
- Integration of web-based frameworks enables easy scalability and maintenance.

These findings highlight the effectiveness of the proposed system in addressing the challenges of modern metro ticketing.

9.3 SIGNIFICANCE OF THE RESEARCH

This research is important because it helps advance both theoretical knowledge and practices in the relevant sphere.

In academic terms, the research proves the applicability of deep learning and computer vision approaches in addressing existing problems. Thus, it may be used as an example of integration between facial recognition technology and transaction systems.

On an industrial level, the proposed system is useful for metro companies that are looking to streamline their work and provide better services for their clients. Using similar systems may help save money and increase the level of safety.

Moreover, the research may contribute to the development of smart transport systems, which are vital for sustainable urban planning. By allowing contactless and automated payment of tickets, the system helps make traveling easier.

9.4 PRACTICAL APPLICATIONS

The proposed system has several practical applications beyond metro ticketing.

- **Public Transportation Systems:** Can be implemented in buses, trains, and airports for automated passenger verification.
- **Smart City Infrastructure:** Can be integrated with services such as parking, toll collection, and access control.
- **Retail and Payment Systems:** Can be adapted for contactless payment solutions in retail environments.
- **Security Systems:** Can be used for access control in restricted areas such as offices and government facilities.
- **Event Management:** Can be used for automated entry management in large events and exhibitions.

These applications demonstrate the versatility and scalability of the proposed system.

9.5 LIMITATIONS OF THE STUDY

Despite its advantages, the system has certain limitations that must be addressed for real-world deployment.

9.5.1 Environmental Factors

The system's performance is affected by lighting conditions. Poor lighting or extreme brightness can reduce detection and recognition accuracy.

9.5.2 Occlusion Issues

The presence of masks, sunglasses, or other accessories can partially block facial features, affecting recognition performance.

9.5.3 Hardware Dependency

The system requires high-quality cameras and sufficient computational resources for real-time processing.

9.5.4 Privacy Concerns

The use of biometric data raises privacy and ethical concerns. Proper data protection measures must be implemented.

9.5.5 Scalability Challenges

Handling large datasets and high user traffic requires efficient database management and optimized algorithms.

9.6 FUTURE ENHANCEMENTS

The proposed system can be further improved by incorporating advanced technologies and features.

9.6.1 Integration with Real Banking Systems

The current system uses a dummy payment mechanism. Future work can integrate real-time banking APIs for actual fare deduction.

9.6.2 Mask Detection and Recognition

Enhancing the system to recognize faces even with masks will improve performance in real-world scenarios.

9.6.3 Cloud-Based Deployment

Deploying the system on cloud platforms can improve scalability, storage, and accessibility.

9.6.4 Mobile Application Integration

Developing a mobile application can provide additional features such as ticket history and notifications.

9.6.5 Edge Computing

Using edge devices for processing can reduce latency and improve real-time performance.

9.6.6 Multi-Factor Authentication

Combining facial recognition with other authentication methods such as OTP can enhance security.

9.6.7 Large-Scale Deployment

Testing the system in real metro environments with thousands of users can validate scalability.

9.6.8 AI Model Optimization

Improving model efficiency can reduce computational requirements and increase speed.

9.7 BROADER IMPACT OF THE RESEARCH

Implications of the proposed research work on society and technology include several factors.

Firstly, the integration of a contactless ticketing solution can contribute towards improving the overall public transportation service by reducing time wastage. Moreover, the technology can ensure safety by eliminating direct human interaction.

From an economic viewpoint, the proposed technology will cut down the cost of manual operations and provide an innovative way for further developments.

In terms of technology, the implementation of such a solution would contribute towards advancing technology, particularly in the field of artificial intelligence.

Social implications of using such a technology include greater convenience for users.

9.8 FINAL REMARKS

The successful implementation of the suggested system is an example of how facial recognition technology may revolutionize metro ticketing systems in the future. The

use of advanced technology and internet services allows creating an effective solution to today's transportation problems.

Even though there may be some drawbacks to the application, they can be fixed in the future through more research in the area. The project has set a great basis for further development of transportation technologies.

To conclude, the suggested project is a step towards making fully automated metro ticketing systems in the future.

- A brief summary, just a few paragraphs, of your key findings, related back to what you expected to see (essential);
- The conclusions which you have drawn from your research (essential);
- Why your research is important for researchers and practitioners (essential);
- Recommendations for future research (strongly recommended, verging on essential);
- Recommendations for practitioners (strongly recommended in management and business courses and some other areas, so check with your supervisor whether this will be expected); and a final paragraph rounding off your dissertation or thesis.

In any research-oriented work, references and citations play a crucial role in establishing the credibility and reliability of the study. Citations are used within the main body of the report to acknowledge the original sources of ideas, theories, methodologies, and results that have informed the research. They allow the reader to trace the origin of information, verify claims, and gain deeper insight into the background literature?.

References, on the other hand, are listed at the end of the dissertation or thesis. They provide the complete bibliographic details of all the sources cited in the text. A well-structured reference list demonstrates the breadth and depth of the literature consulted during the research process. It also helps future researchers to locate relevant works more efficiently?.

In this report, references have been cited in accordance with standard academic practice. Each in-text citation corresponds to a detailed entry in the reference list.

Sources include journal papers, conference proceedings, books, technical reports, and reputable online resources. The citation style followed (e.g., IEEE, APA, or any style recommended by the institute) ensures uniformity and consistency throughout the document?.

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Appendix A- Sample Code

```
Backend Code

from django.shortcuts import render, redirect
from django.contrib.auth.models import User
from django.contrib.auth import login
from django.http import JsonResponse
from .models import Customer, Transaction
from deepface import DeepFace
import cv2
import numpy as np
import base64
import io
from PIL import Image
from django.contrib import messages
from django.core.files.base import ContentFile
import face_recognition
import re

# Create your views here.
def home(request):
    # This will load a template file called home.html
    return render ( request, 'home.html' )


def checkout(request):
    error = ""
    success = ""

    if request.method == "POST":
        mobile = request.POST.get("mobile_number")
        amount = float(request.POST.get("amount"))
        face_data = request.POST.get("face_data")

        try:
            customer = Customer.objects.get(mobile_number=mobile)
        except Customer.DoesNotExist:
            error = "Mobile number not registered."
            return render(request, "checkout.html", {"error": error})

        if not face_data:
            error = "Please capture your face."
            return render(request, "checkout.html", {"error": error})

        # Convert base64 to image
        face_bytes = base64.b64decode(face_data.split(',')[1])
```

```

img = Image.open(io.BytesIO(face_bytes))
img = np.array(img)

try:
    # Get embedding of live image
    live_embedding = DeepFace.represent(img, model_name='Facenet')[0]["embedding"]
    live_embedding = np.array(live_embedding, dtype=np.float32)

    # Get saved embedding
    saved_embedding = np.frombuffer(customer.face_embedding, dtype=np.float32)

    # Compare embeddings
    distance = np.linalg.norm(saved_embedding - live_embedding)
    if distance > 0.6: # Threshold for match
        error = "Face not recognized. Transaction denied."
        return render(request, "checkout.html", {"error": error})

except Exception as e:
    error = f"Face verification failed: {str(e)}"
    return render(request, "checkout.html", {"error": error})

# Dummy payment
if customer.balance >= amount:
    customer.balance -= amount
    customer.save()
    Transaction.objects.create(sender=customer, receiver=customer, amount=amount, status=
    success = f"Payment of {amount} successful! Remaining balance: {customer.balance}"
else:
    Transaction.objects.create(sender=customer, receiver=customer, amount=amount, status=
    error = "Insufficient balance."

return render(request, "checkout.html", {"error": error, "success": success})

*****

#++++++

#Updated code

# ----- SIGNUP -----

def signup(request):
    if request.method == "POST":
        first_name = request.POST.get("first_name")
        last_name = request.POST.get("last_name")
        identity_number = request.POST.get("identity_number") # Aadhaar
        mobile_number = request.POST.get("mobile_number")
        bank_name = request.POST.get("bank_name")
        account_number = request.POST.get("account_number")
        face_image_data = request.POST.get("face_image")

```



```

# decode base64 -> numpy image
header, img_b64 = face_image_data.split(";base64,")
decoded = base64.b64decode(img_b64)
np_img = np.frombuffer(decoded, np.uint8)
img = cv2.imdecode(np_img, cv2.IMREAD_COLOR)

# extract face encodings
face_enc = face_recognition.face_encodings(img)
if not face_enc:
    return JsonResponse({"status": "error", "message": "No face detected"})

encoding = face_enc[0].tobytes()

# create customer
customer = Customer.objects.create(
    first_name=first_name,
    last_name=last_name,
    identity_number=identity_number,
    mobile_number=mobile_number,
    bank_name=bank_name,
    account_number=account_number,
    face_encoding=encoding
)

return render(request, "signup.html", {"success": True})

return render(request, "signup.html")

#return JsonResponse({"status": "success", "message": "Signup successful"})

#return render(request, "signup.html")

'''def signup(request):
    if request.method == "POST":
        first_name = request.POST.get("first_name")
        last_name = request.POST.get("last_name")
        identity_proof = request.POST.get("identity_proof")
        identity_number = request.POST.get("identity_number")
        mobile_number = request.POST.get("mobile_number")
        bank_name = request.POST.get("bank_name")
        account_number = request.POST.get("account_number")
        face_image_data = request.POST.get("face_image")

```

```

#         check if face image exists
if not face_image_data:
    return JsonResponse({"success": False, "reason": "No face image captured"})

try:
    # decode face image
    header, img_b64 = face_image_data.split(';base64,')
    decoded = base64.b64decode(img_b64)
    img = face_recognition.load_image_file(io.BytesIO(decoded))

    encodings = face_recognition.face_encodings(img)
    if not encodings:
        return JsonResponse({"success": False, "reason": "No face detected"})

    face_encoding = np.array(encodings[0]).tobytes()
except Exception as e:
    import traceback; traceback.print_exc()
    return JsonResponse({"success": False, "reason": "Face processing failed"})

#         Save customer
Customer.objects.create(
    first_name=first_name,
    last_name=last_name,
    identity_proof=identity_proof,
    identity_number=identity_number,
    mobile_number=mobile_number,
    bank_name=bank_name,
    account_number=account_number,
    face_encoding=face_encoding
)

return JsonResponse({"success": True, "redirect": "/home/"})

return render(request, "signup.html")
'''

# ----- SIGNIN -----
def signin(request):
    if request.method == "POST":
        step = request.POST.get("step")
        mobile = request.POST.get("mobile_number")

    # Step 1: check mobile
    if step == "check_mobile":
        try:
            customer = Customer.objects.get(mobile_number=mobile)
            aadhaar = getattr(customer, "identity_number", "")
            masked = "XXXX XXXX " + aadhaar[-4:] if len(aadhaar) >= 4 else "XXXX XXXX"

```

```

        return JsonResponse({"exists": True, "aadhaar": masked})
    except Customer.DoesNotExist:
        return JsonResponse({"exists": False})

# Step 2: validate face
if step == "validate_face":
    try:
        customer = Customer.objects.get(mobile_number=mobile)
    except Customer.DoesNotExist:
        return JsonResponse({"face_valid": False, "reason": "Customer not found"})

    face_image_data = request.POST.get("face_image")
    if not face_image_data:
        return JsonResponse({"face_valid": False, "reason": "No face image provided"})

    try:
        # decode capture
        header, img_b64 = face_image_data.split(";base64,")
        decoded = base64.b64decode(img_b64)
        np_img = np.frombuffer(decoded, np.uint8)
        img = cv2.imdecode(np_img, cv2.IMREAD_COLOR)

        captured_enc = face_recognition.face_encodings(img)
        if not captured_enc:
            return JsonResponse({"face_valid": False, "reason": "No face detected in cap

        if not customer.face_encoding:
            return JsonResponse({"face_valid": False, "reason": "No stored face data"})

        stored_enc = np.frombuffer(customer.face_encoding, dtype=np.float64)

        match = face_recognition.compare_faces([stored_enc], captured_enc[0], tolerance=0.5)
        if match[0]:
            return JsonResponse({"face_valid": True})
        else:
            return JsonResponse({"face_valid": False, "reason": "Face mismatch"})

    except Exception as e:
        import traceback; traceback.print_exc()
        return JsonResponse({"face_valid": False, "reason": "Server error during face ch

# Step 3: make payment (dummy)
if step == "make_payment":
    return JsonResponse({"payment": "success"})

return render(request, "signin.html")

def success(request):

```

```
return render(request, 'success.html')
```

Front end Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>FacePay Registration</title>
<link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;500;700&display=swap" rel="stylesheet">
<style>
    body {
        font-family: 'Roboto', sans-serif;
        background: linear-gradient(135deg, #667eea, #764ba2);
        display: flex;
        justify-content: center;
        align-items: center;
        min-height: 100vh;
        margin: 0;
    }

    .container {
        h1{
            text-align: center;
        }
        background: rgba(141, 92, 190, 0.95);
        padding: 40px;
        border-radius: 20px;
        box-shadow: 0 15px 35px rgba(0,0,0,0.2);
        width: 90%;
        max-width: 1200px;
        min-height: 90vh;
        display: flex;
        flex-direction: column;
        justify-content: center;
    }

    h2 {
        margin-bottom: 30px;
        color: #333;
        font-weight: 700;
        font-size: 32px;
        text-align: center;
    }

    form {
        display: grid;
    }
```

```

        grid-template-columns: 1fr 1fr;
        gap: 20px;
    }

    input, select, button {
        width: 100%;
        padding: 15px;
        border-radius: 10px;
        border: 1px solid #ccc;
        font-size: 18px;
    }

    input:focus, select:focus {
        border-color: #667eea;
        box-shadow: 0 0 8px rgba(102,126,234,0.5);
        outline: none;
    }

    button {
        cursor: pointer;
        font-weight: 600;
    }

    button#captureBtn {
        background: #48bb78;
        color: #fff;
        border: none;
        grid-column: span 2;
    }

    button#captureBtn:hover { background: #38a169; }

    button#submitBtn {
        background: #667eea;
        color: #fff;
        border: none;
        grid-column: span 2;
    }

    button#submitBtn:hover { background: #5a67d8; }

    .video-wrapper {
        grid-column: span 2;
        margin: 20px auto;
        width: 400px;
        height: 400px;
        border-radius: 50%;
        overflow: hidden;
        border: 5px solid #48bb78;
        display: none;
    }

```

```

#video {
    width: 100%;
    height: 100%;
    object-fit: cover;
}

#canvas { display: none; }

.error { color: #e53e3e; font-size: 16px; grid-column: span 2; text-align: center; }
.success { color: #38a169; font-size: 16px; grid-column: span 2; text-align: center; }

@media (max-width: 768px) {
    form { grid-template-columns: 1fr; }
    .video-wrapper { width: 300px; height: 300px; }
}

.success-message {
    margin-top: 30px;
    font-size: 20px;
    font-weight: bold;
    color: #38a169;
    background: #e6fffa;
    padding: 15px;
    border-radius: 10px;
}
}
</style>
</head>
<body>

<div class="container">
    <b>
        <u>
            <h1> We are happy to Welcoming You</h1>
        </u>
    </b>
    <b>
        <h2>Face Pay<=>Smart Way</h2>
    </b>
    <form method="POST" enctype="multipart/form-data">
        {% csrf_token %}
        <input type="text" name="first_name" placeholder="First Name" required>
        <input type="text" name="last_name" placeholder="Last Name" required>

        <select id="identity_proof" name="identity_proof" required>
            <option value="">Select Proof of Identity</option>
            <option value="aadhar">Aadhar Card</option>
            <option value="pan">PAN Card</option>
            <option value="passport">Passport</option>
        </select>
    </form>
</div>

```

```

        <option value="driving_license">Driving License</option>
    </select>

    <input type="text" id="identity_number" name="identity_number" placeholder="Enter Identity Number" />

    <input type="text" id="mobile_number" name="mobile_number" placeholder="Mobile Number (10 Digits)" />

    <select id="bank_name" name="bank_name" required>
        <option value="">Please Select your bank</option>
        <option value="SBI">State bank Of India</option>
        <option value="HDFC">HDFC India</option>
        <option value="ICICI">ICICI Bank</option>
        <option value="Axis">Axis Bank</option>
        <option value="Canara">Canara Bank</option>
        <option value="Yes">Yes Bank India Private Limited</option>
        <option value="India">Indian Bank</option>
        <option value="BOB">Bank of Baroda</option>
        <option value="Union">Union Bank</option>
        <option value="PNJ">Punjab National Bank</option>
    </select>

    <!-- Searchable Bank Dropdown
    <input list="bankList" id="bank_name" name="bank_name" placeholder="Please select your Bank" />
    <datalist id="bankList">
        <option value="SBI"></option>
        <option value="HDFC"></option>
        <option value="ICICI"></option>
        <option value="Axis"></option>
        <option value="Canara"></option>
        <option value="PNB"></option>
    </datalist>
-->

    <!-- Readonly Account Number -->
    <input type="text" id="account_number" name="account_number" placeholder="Account Number" />

    <div class="video-wrapper">
        <video id="video" autoplay></video>
    </div>
    <canvas id="canvas" width="400" height="400"></canvas>

    <button type="button" id="captureBtn">Start Camera</button>
    <input type="hidden" name="face_image" id="faceImage">

    <button type="submit" id="submitBtn">SIGNUP</button>

    {% if error %}<div class="error">{{ error }}</div>{% endif %}
    {% if success %}
    <div class="success-message">

```

```

        <h1>THANK YOU FOR USING OUR APPLICATION</h1>
        <h2>We providing 100% security for your data</h2>
    </div>
    <script>
        setTimeout(function() {
            window.location.href = "{% url 'home' %}";
        }, 2000);
    </script>
    {% endif %}
</form>
</div>

<script>
/* ----- Face Capture ----- */
const video = document.getElementById('video');
const canvas = document.getElementById('canvas');
const faceImage = document.getElementById('faceImage');
const captureBtn = document.getElementById('captureBtn');
const videoWrapper = document.querySelector('.video-wrapper');

let stream = null;
let cameraStarted = false;

captureBtn.addEventListener('click', async () => {
    if (!cameraStarted) {
        stream = await navigator.mediaDevices.getUserMedia({ video: true });
        video.srcObject = stream;
        videoWrapper.style.display = 'block';
        captureBtn.textContent = "Capture Face";
        cameraStarted = true;
    } else {
        canvas.getContext('2d').drawImage(video, 0, 0, canvas.width, canvas.height);
        const dataURL = canvas.toDataURL('image/png');
        faceImage.value = dataURL;

        stream.getTracks().forEach(track => track.stop());
        videoWrapper.style.display = 'none';
        alert("Face captured successfully!");
        captureBtn.textContent = "Start Camera";
        cameraStarted = false;
    }
});

/* ----- Aadhaar Validation ----- */
document.getElementById('identity_proof').addEventListener('change', function () {
    const identityInput = document.getElementById('identity_number');
    if (this.value === 'aadhar') {
        identityInput.setAttribute('pattern', '\\d{12}');
        identityInput.setAttribute('maxlength', '12');
    }
});

```



```

        identityInput.setAttribute('title', 'Aadhaar must be exactly 12 digits');
        identityInput.placeholder = "Enter Aadhaar Number";
    } else {
        identityInput.removeAttribute('pattern');
        identityInput.removeAttribute('maxlength');
        identityInput.removeAttribute('title');
        identityInput.placeholder = "Enter Identity Number";
    }
});

/* ----- Auto-generate Account Number ----- */
function generateAccountNumber(bank, mobile) {
    if (!mobile || mobile.length < 4) return "";
    let suffix = mobile.slice(-4); // last 4 digits of mobile
    switch(bank.toLowerCase()) {
        case "sbi": return "10" + suffix + "001";
        case "hdfc": return "20" + suffix + "002";
        case "icici": return "30" + suffix + "003";
        case "axis": return "40" + suffix + "004";
        default: return "99" + suffix + "000";
    }
}

const bankInput = document.getElementById("bank_name");
const mobileInput = document.getElementById("mobile_number");
const accInput = document.getElementById("account_number");

function updateAccountNumber() {
    const bank = bankInput.value;
    const mobile = mobileInput.value;
    accInput.value = generateAccountNumber(bank, mobile);
}

bankInput.addEventListener("input", updateAccountNumber);
mobileInput.addEventListener("input", updateAccountNumber);
</script>

</body>
</html>

```

Appendix B- Publication details

- Apruzzese, G., Laskov, P., Montes de Oca, E., Mallouli, W., Brdalo Rapa, L., Grammatopoulos, A. V., & Di Franco, F. (2023). The role of machine learning in cybersecurity. *Digital Threats: Research and Practice*, 4(1), 1-38.
- Kumar, S., Gupta, U., Singh, A. K., & Singh, A. K. (2023). AI: Revolutionizing cyber security in the Digital Era. *J. Comput. Mech. Manag*, 2(3), 31-42.